

# Q1. Explain the components of a Data Communication System. Discuss analog and digital signals and the role of encoders and modems.

## 1. Introduction

**Data communication** is the process of exchanging data between two or more devices through a communication channel or transmission medium. An effective data communication system must ensure **delivery, accuracy, timeliness, and jitter control** where applicable.

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## 2. Components of a Data Communication System

A data communication system consists of the following fundamental components:

### 1. Message

The **message** is the information to be communicated. It may consist of:

- Text
- Numerical data
- Audio
- Images
- Video

### 2. Sender

The **sender** is the source device that generates and transmits the message.

Examples include a computer, mobile device, server, or sensor.

### 3. Receiver

The **receiver** is the destination device that accepts and processes the transmitted message.

### 4. Transmission Medium

The **transmission medium** is the physical or logical channel through which the data propagates.

Examples:

- Twisted-pair cable
- Coaxial cable
- Optical fibre
- Radio waves

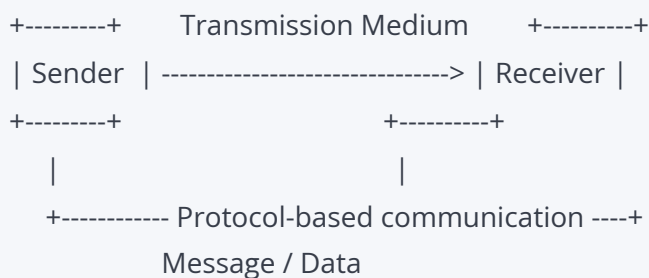
## 5. Protocol

A **protocol** is a set of communication rules governing data exchange between network entities.

A protocol generally specifies:

- **Syntax** – Structure and format of data
- **Semantics** – Meaning of individual fields and control information
- **Timing** – Data transmission rate and synchronization

### Basic communication model



## 3. Analog and Digital Signals

### Analog Signal

An **analog signal** is a continuous-time signal whose amplitude can vary continuously over a specified range.

Its primary characteristics are:

- **Amplitude**
- **Frequency**
- **Phase**

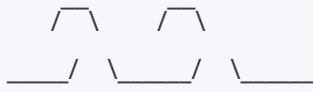
Analog signals can be affected significantly by **noise, attenuation, and distortion**.

Examples:

- Traditional voice signals

- Radio signals

#### Representation

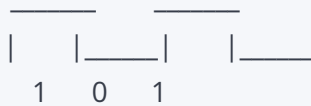


#### Digital Signal

A **digital signal** represents information using a finite number of discrete signal levels. In computer networks, binary digital signals commonly represent:

- Logic 0
- Logic 1

#### Representation



Digital communication provides improved noise immunity because the receiver can regenerate the original signal at intermediate stages.

#### Comparison

| Parameter      | Analog Signal       | Digital Signal         |
|----------------|---------------------|------------------------|
| Nature         | Continuous          | Discrete               |
| Values         | Infinite            | Finite                 |
| Noise immunity | Lower               | Higher                 |
| Regeneration   | Difficult           | Easier                 |
| Error control  | More complex        | More effective         |
| Representation | Continuous waveform | Binary/discrete levels |

## 4. Encoder

An **encoder** converts information or data into a suitable coded representation for transmission over a communication channel.

In digital communication, encoding converts binary data into an appropriate signal format according to the characteristics of the transmission medium.

The primary functions of an encoder include:

- Signal representation
  - Data formatting
  - Supporting synchronization
  - Facilitating reliable transmission
- 

## 5. Modem

The term **modem** is derived from:

*MOdulator + DEModulator*

A modem enables communication by performing conversion between digital information and signals suitable for the transmission channel.

### Modulation

At the transmitting side:

Digital Data   Modulator   Modulated Signal

### Demodulation

At the receiving side:

Received Signal   Demodulator   Digital Data

### Overall operation

Computer   Modem   Communication Channel   Modem   Computer  
                  Modulation                                  Demodulation

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## Q2. Compare Circuit Switching, Message Switching, and Packet Switching with suitable diagrams and applications.

### 1. Introduction

**Switching** is the technique used to transfer information from a source to a destination through an interconnected network. Switching systems determine the path and method used for forwarding data through intermediate nodes.

The major switching techniques are:

1. Circuit Switching
  2. Message Switching
  3. Packet Switching
- 

### 2. Circuit Switching

In **circuit switching**, a dedicated end-to-end communication path is established before the actual data transfer begins.

The communication process consists of three phases:

#### 1. Connection Establishment

A dedicated circuit is established between source and destination.

#### 2. Data Transfer

Data is transmitted over the reserved communication path.

#### 3. Connection Termination

The circuit is released after communication is completed.

#### Architecture

Source ===== Switch ===== Switch ===== Destination  
Dedicated End-to-End Path

#### Characteristics

- Dedicated path
- Resources reserved for the communication session
- Continuous data transmission after setup

- Predictable communication characteristics
- Setup phase required

#### Application

Traditional **telephone networks** are the classical example of circuit switching.

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### 3. Message Switching

In **message switching**, the entire message is transmitted from one intermediate node to another using the **store-and-forward** principle.

A dedicated path is not established.

#### Operation

|        |          |          |          |             |
|--------|----------|----------|----------|-------------|
| Source | [Node 1] | [Node 2] | [Node 3] | Destination |
|        | Store    | Store    | Store    |             |
|        | Forward  | Forward  | Forward  |             |

Each intermediate node:

1. Receives the complete message.
2. Stores it temporarily.
3. Examines the destination address.
4. Selects an appropriate outgoing link.
5. Forwards the complete message.

#### Characteristics

- No dedicated path
- Entire message is stored before forwarding
- Requires significant buffer storage
- Variable and potentially high delay
- Efficient sharing of network resources

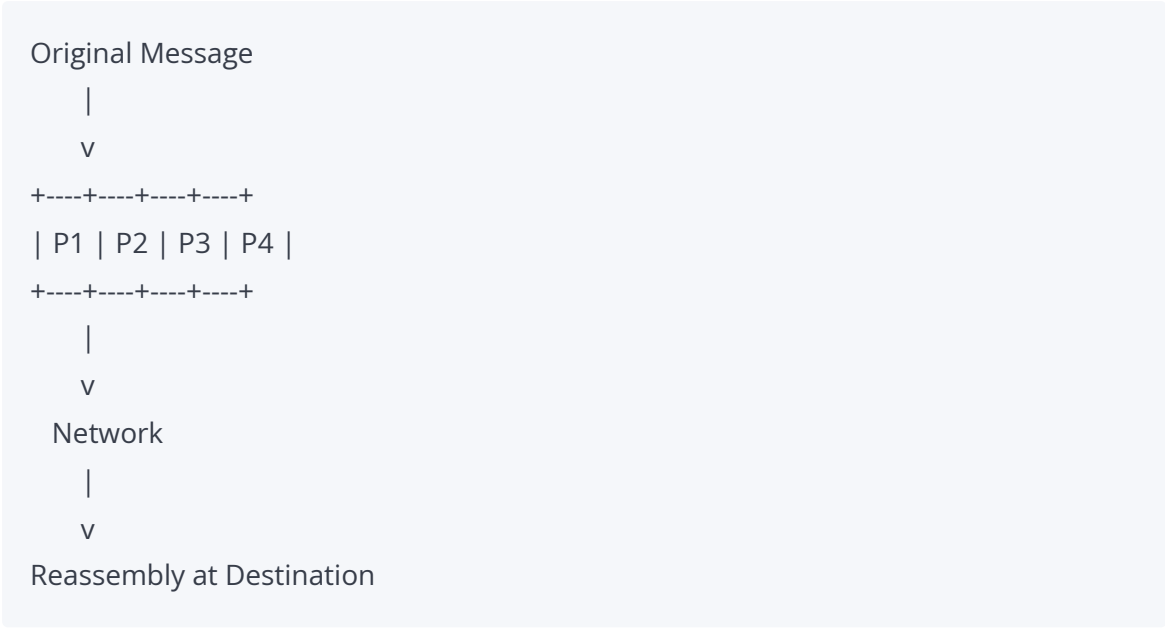
#### Application

Store-and-forward communication systems and early telegraph networks.

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### 4. Packet Switching

In **packet switching**, a large message is divided into smaller units called **packets**. Each packet contains data and control information required for delivery.



**Characteristics**

- No dedicated path is required.
- Multiple users share the available network resources.
- Messages are transmitted as smaller packets.
- Efficient bandwidth utilization.
- Packets may experience variable delays.
- Destination reassembles packets into the original message.

**Application**

The **Internet** is based primarily on packet-switched communication.

**5. Comparative Analysis**

| Parameter            | Circuit Switching      | Message Switching | Packet Switching       |
|----------------------|------------------------|-------------------|------------------------|
| Communication path   | Dedicated              | Not dedicated     | Not dedicated          |
| Unit of transmission | Continuous data stream | Complete message  | Packet                 |
| Connection setup     | Required               | Not required      | Generally not required |

|                              |                           |                  |                    |
|------------------------------|---------------------------|------------------|--------------------|
| Storage at intermediate node | Minimal                   | Complete message | Individual packets |
| Resource allocation          | Reserved                  | Shared           | Shared             |
| Delay                        | Low after setup           | High             | Variable           |
| Efficiency                   | Lower during idle periods | Better           | High               |
| Example                      | Telephone network         | Telegraph system | Internet           |

### Q3. Explain different Network Topologies. Compare their advantages and disadvantages.

#### 1. Introduction

A **network topology** defines the physical or logical arrangement of nodes and communication links within a computer network.

The commonly used topologies are:

1. Bus
2. Star
3. Ring
4. Mesh
5. Tree

#### 2. Bus Topology

In a **bus topology**, all network devices share a common communication medium known as the **backbone**.

```
[Node] ---- [Node] ---- [Node] ---- [Node]
          Shared Backbone
```

##### Advantages

- Simple architecture
- Low implementation cost

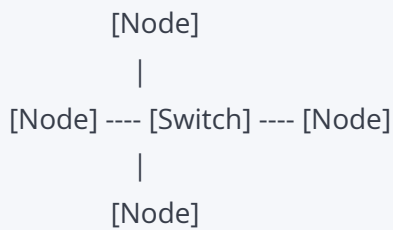


#### Disadvantages

- Backbone failure can affect the entire network
  - Difficult fault isolation
- 

### 3. Star Topology

In a **star topology**, every node has a dedicated connection to a central device such as a hub or switch.



#### Advantages

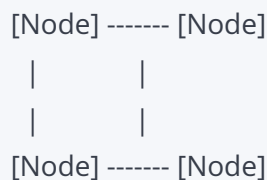
- Easy installation and management
- Simplified fault detection

#### Disadvantages

- Central device represents a potential single point of failure
  - Requires more cabling than a bus topology
- 

### 4. Ring Topology

In a **ring topology**, every node is connected to two adjacent nodes, forming a closed communication loop.



#### Advantages

- Structured and orderly communication
- Each node has a defined position

### Disadvantages

- A link or node failure can disrupt communication
  - Fault diagnosis can be difficult
- 

## 5. Mesh Topology

In a **mesh topology**, nodes are interconnected through multiple links.

### Types

- **Full Mesh** – Every node is connected directly to every other node.
- **Partial Mesh** – Only selected nodes have multiple direct connections.



### Advantages

- High reliability
- Multiple alternative paths

### Disadvantages

- High cabling and hardware requirements
- Complex installation

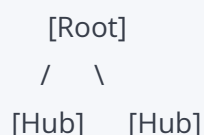
For a full mesh containing **n nodes**, the number of physical links is:

$$n(n - 1) / 2$$

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## 6. Tree Topology

A **tree topology** follows a hierarchical structure. It connects multiple groups of nodes through a root or higher-level network device.



```
 / \  / \
[N1] [N2] [N3] [N4]
```

#### Advantages

- Highly scalable
- Suitable for large networks

#### Disadvantages

- Failure of higher-level devices can affect multiple nodes
- More complex than bus and star architectures

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## Q4. Explain the OSI Reference Model in detail. Compare it with the TCP/IP Model.

### 1. Introduction

The **OSI (Open Systems Interconnection) Reference Model** is a conceptual framework developed by ISO to standardize the functions involved in network communication. It divides the communication process into **seven distinct layers**, each performing specific functions and providing services to the layer above it.

The **TCP/IP model** is a practical protocol architecture used as the basis of the Internet.

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### 2. OSI Reference Model

```
+-----+
| 7. Application  |
+-----+
| 6. Presentation |
+-----+
| 5. Session      |
+-----+
| 4. Transport    |
+-----+
| 3. Network      |
+-----+
| 2. Data Link    |
```

```
+-----+
| 1. Physical |
+-----+
```

---

### 3. Physical Layer

The **Physical Layer** is responsible for transmitting raw bits over the physical communication medium.

#### Functions

- Transmission and reception of bits
- Definition of electrical and mechanical interfaces
- Data rate specification
- Bit synchronization
- Physical transmission media specification

#### Data Unit

#### Bit

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### 4. Data Link Layer

The **Data Link Layer** provides node-to-node delivery and improves the reliability of communication over a physical link.

#### Functions

- Framing
- Physical addressing
- Error detection and correction
- Flow control
- Medium access control

#### Data Unit

#### Frame

The Data Link Layer is commonly divided into:

- **Logical Link Control (LLC)**
- **Media Access Control (MAC)**

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## 5. Network Layer

The **Network Layer** is responsible for delivering packets from the source network to the destination network.

### Functions

- Logical addressing
- Routing
- Packet forwarding
- Path determination
- Congestion management

### Data Unit

### Packet

**Example:** Internet Protocol (IP)

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## 6. Transport Layer

The **Transport Layer** provides end-to-end communication between processes running on source and destination systems.

### Functions

- Segmentation and reassembly
- End-to-end delivery
- Flow control
- Error control
- Multiplexing and demultiplexing

### Data Unit

### Segment

### Examples:

- TCP
  - UDP
-

## 7. Session Layer

The **Session Layer** establishes, manages, and terminates communication sessions between applications.

### Functions

- Session establishment
  - Session maintenance
  - Session termination
  - Dialog control
  - Synchronization and checkpointing
- 

## 8. Presentation Layer

The **Presentation Layer** is responsible for the representation and interpretation of data.

### Functions

- Data translation
- Data format conversion
- Encryption and decryption
- Compression and decompression

Its purpose is to ensure that data generated by one system can be correctly interpreted by another system.

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## 9. Application Layer

The **Application Layer** provides network services directly to end-user applications.

### Functions

- File transfer
- Electronic mail
- Remote access
- Web communication
- Name resolution

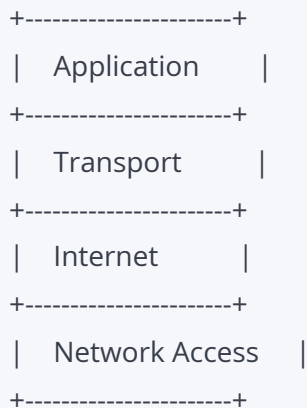
### Examples

- HTTP

- FTP
  - SMTP
  - DNS
- 

## 10. TCP/IP Model

The TCP/IP model is generally represented using **four layers**:



### 1. Application Layer

Combines the functionality of the OSI:

- Application Layer
- Presentation Layer
- Session Layer

### 2. Transport Layer

Provides process-to-process communication.

#### Protocols:

- TCP
- UDP

### 3. Internet Layer

Responsible for:

- Logical addressing
- Routing
- Packet delivery

### Protocols:

- IP
- ICMP

## 4. Network Access Layer

Handles communication over the physical network and includes functions corresponding broadly to:

- Data Link Layer
  - Physical Layer
- 

## Conclusion

The OSI model provides a **layered conceptual framework** for understanding network communication, whereas TCP/IP provides the **practical protocol architecture** used by modern Internet-based networks. The layered approach improves modularity, interoperability, and protocol design.

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## Q5. Explain Error Detection and Error Correction Techniques. Discuss CRC and Hamming Code with suitable examples.

### 1. Introduction

During data transmission, errors may occur due to:

- Noise
- Interference
- Signal attenuation
- Transmission impairments

The Data Link Layer uses **error detection and error correction mechanisms** to improve the reliability of transmitted data.

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### 2. Error Detection and Error Correction



## Error Detection

Error detection enables the receiver to determine whether the received data has been corrupted.

Common techniques include:

- Parity checking
- Checksum
- Cyclic Redundancy Check (CRC)

## Error Correction

Error correction enables the receiver to identify and correct errors without necessarily requesting retransmission.

A common technique is:

- Hamming Code
- 

## 3. Cyclic Redundancy Check (CRC)

CRC is a powerful error-detection technique based on **modulo-2 polynomial division**.

A predetermined binary sequence called the **generator** is used by both sender and receiver.

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### CRC Working Procedure

Assume:

- Dataword =  $D$
- Generator =  $G$
- Generator length =  $n$

#### Step 1: Append Zeros

If the generator contains  $n$  bits, append  **$n - 1$  zeros** to the original dataword.

Dataword     $D000\dots0$

## Step 2: Perform Modulo-2 Division

Divide the modified dataword by the generator using **modulo-2 arithmetic**, where subtraction is equivalent to XOR.

## Step 3: Obtain the Remainder

The remainder contains  $n - 1$  bits.

## Step 4: Form the Codeword

The remainder is appended to the original dataword:

Codeword = Dataword + CRC Remainder

## Step 5: Receiver Verification

The receiver divides the received codeword by the same generator.

Received Codeword

|

v

Modulo-2 Division using Generator

|

v

Remainder

- **Remainder = 0:** No error detected
- **Non-zero remainder:** Error detected

## Advantages of CRC

- High error-detection capability
- Efficient hardware implementation

## Limitation

CRC generally **detects errors but does not correct them**.

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## 4. Hamming Code

**Hamming Code** is a forward error correction technique capable of identifying and correcting **single-bit errors**.

It uses additional bits called **parity bits** or **redundant bits**.

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### Number of Parity Bits

For **m** data bits and **r** parity bits:

$$2^r \geq m + r + 1$$

The parity bits are placed at positions:

$$1, 2, 4, 8, 16, \dots$$

These are powers of 2.

---

### Working of Hamming Code

Assume four data bits are transmitted.

#### Step 1: Determine parity bits

For:

$$m = 4$$

We need:

$$2^r \geq 4 + r + 1$$

For **r = 3** :

$$2^3 = 8 \geq 8$$

Therefore, **three parity bits** are required.

---

#### Step 2: Assign bit positions

| Position | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
|----------|----|----|----|----|----|----|----|
| Bit      | P1 | P2 | D1 | P4 | D2 | D3 | D4 |

Parity bits are placed at positions **1, 2, and 4**.

---

### Step 3: Parity Checking

Each parity bit checks a specific combination of bit positions.

| Parity Bit | Positions Checked |
|------------|-------------------|
| P1         | 1, 3, 5, 7        |
| P2         | 2, 3, 6, 7        |
| P4         | 4, 5, 6, 7        |

The receiver checks these parity groups. The result forms a **syndrome**, which identifies the erroneous bit position.

---

### Example of Error Correction

Assume the received codeword is:

1010011

If the parity-check syndrome is:

$101_2 = 5$

Then the error is located at **bit position 5**.

The receiver corrects the error by inverting bit 5.

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## Q6. Explain the Sliding Window Protocol and discuss Stop-and-Wait, Go-Back-N ARQ, and Selective Repeat ARQ.

### 1. Introduction

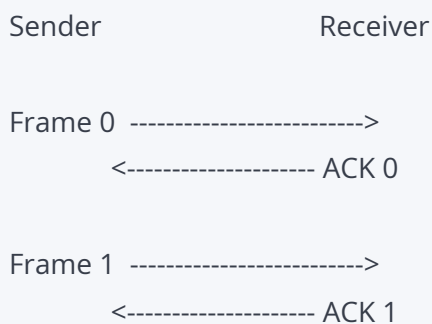
**Flow control** regulates the amount of data transmitted by a sender so that the receiver is not overwhelmed. **Error control** ensures reliable delivery through acknowledgments, timers, retransmissions, and sequence numbers.

The major protocols include:

1. Stop-and-Wait ARQ
  2. Sliding Window Protocol
  3. Go-Back-N ARQ
  4. Selective Repeat ARQ
- 

## 2. Stop-and-Wait Protocol

In the Stop-and-Wait protocol, the sender transmits **one frame at a time** and waits for an acknowledgment before transmitting the next frame.



### Operation

1. Sender transmits a frame.
2. Sender starts a timer.
3. Receiver receives the frame.
4. Receiver sends an acknowledgment.
5. Sender transmits the next frame after receiving the acknowledgment.
6. If the timer expires, the frame is retransmitted.

### Advantages

- Simple implementation
- Requires minimal buffer space
- Reliable communication

### Limitations

- Low bandwidth utilization
  - Inefficient for high-delay networks
  - Only one outstanding frame is allowed
-

### 3. Sliding Window Protocol

The **Sliding Window Protocol** allows the sender to transmit multiple frames before receiving acknowledgments.

The sender maintains a **send window**, while the receiver maintains a **receive window**.

Sequence Numbers

| ---- Current Window ---- |

0 1 2 3 4 5

[S] [S] [S] [S] [ ] [ ]

As acknowledgments are received, the window moves or **slides forward**.

#### Benefits

- Improved channel utilization
- Supports multiple outstanding frames
- Better performance than Stop-and-Wait
- Supports pipelined transmission

---

### 4. Go-Back-N ARQ

**Go-Back-N ARQ** is a sliding window protocol in which the sender may transmit multiple frames before receiving acknowledgments.

If a frame is lost or damaged, the receiver discards that frame and, in the basic protocol, subsequent out-of-order frames. The sender retransmits the erroneous frame and all subsequent frames.

#### Example

Sender:

Frame 0   Frame 1   Frame 2   Frame 3   Frame 4

Frame 2 lost

Receiver:

Frame 0

Frame 1  
Frame 2  
Frame 3 discarded  
Frame 4 discarded

Sender retransmits:  
Frame 2   Frame 3   Frame 4

### Characteristics

- Multiple frames can be transmitted
- Receiver typically accepts only the next expected frame
- Uses cumulative acknowledgments
- Sender retransmits a sequence of frames after an error

### Advantage

Simpler than Selective Repeat.

### Limitation

May retransmit frames that were already transmitted correctly, reducing efficiency.

---

## 5. Selective Repeat ARQ

**Selective Repeat ARQ** is a sliding window protocol that retransmits only the specific lost or damaged frames.

The receiver can accept and buffer correctly received out-of-order frames.

### Example

Sender:  
Frame 0   Frame 1   Frame 2   Frame 3   Frame 4

Frame 2 lost

Receiver:  
Frame 0  
Frame 1  
Frame 2

Frame 3    Buffered

Frame 4    Buffered

Sender retransmits:

Frame 2 only

### Characteristics

- Individual acknowledgments
- Selective retransmission
- Receiver buffers out-of-order frames
- Higher bandwidth efficiency

### Limitation

- More complex implementation
- Requires larger receiver buffers

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## Q7. Explain Ethernet and IEEE 802.3 LAN Architecture. Discuss Manchester Encoding and Binary Exponential Backoff.

### 1. Introduction

**Ethernet** is a widely used LAN technology that specifies mechanisms for frame transmission and medium access. The IEEE standardized Ethernet under the **IEEE 802.3** family of standards.

The classical Ethernet architecture operates using **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)** in a shared, half-duplex medium.

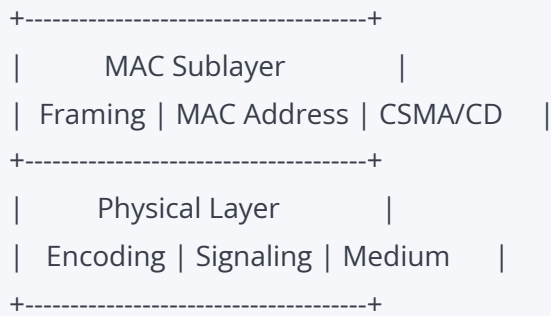
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### 2. IEEE 802.3 Architecture

IEEE 802.3 defines specifications primarily associated with the **Physical Layer** and **MAC sublayer** of the Data Link Layer.

+-----+  
|        LLC Sublayer        |





The two Data Link sublayers are:

### 1. LLC — Logical Link Control

Responsible for providing the interface between the upper layers and the MAC sublayer.

### 2. MAC — Media Access Control

Responsible for controlling access to the shared transmission medium.

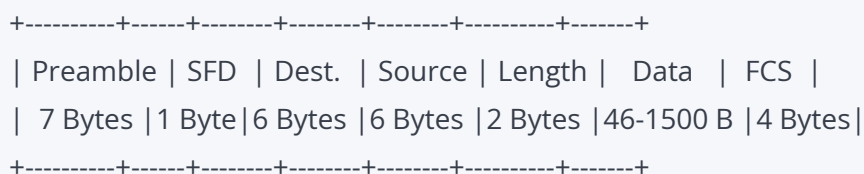
Major MAC functions include:

- Physical addressing
- Frame transmission
- Media access control
- Collision detection in classical shared Ethernet

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## 3. Ethernet Frame Format

A basic IEEE 802.3 frame contains the following fields:



### Preamble

Provides synchronization between the sender and receiver.

### SFD — Start Frame Delimiter

Indicates the actual beginning of the frame.

### **Destination Address**

Contains the **48-bit MAC address** of the intended destination.

### **Source Address**

Contains the **48-bit MAC address** of the sender.

### **Length**

Indicates the size of the data field in IEEE 802.3.

### **Data**

Contains the payload received from upper-layer protocols.

### **FCS — Frame Check Sequence**

Contains a CRC value used for error detection.

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## **4. CSMA/CD**

Classical Ethernet uses **Carrier Sense Multiple Access with Collision Detection**.

### **Carrier Sense**

A station listens to the communication channel before transmitting.

### **Multiple Access**

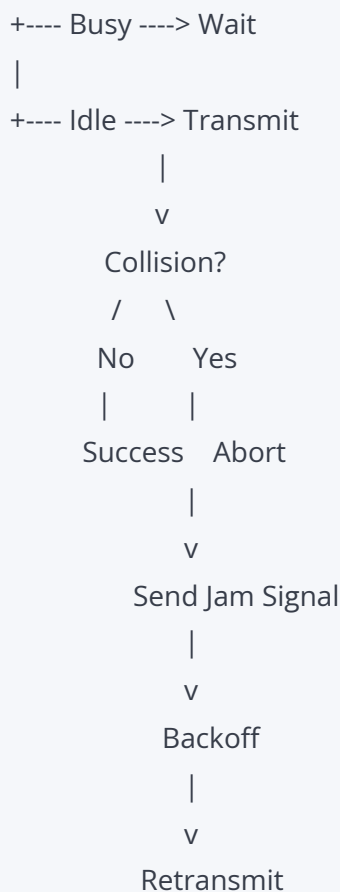
Multiple stations share the same transmission medium.

### **Collision Detection**

A transmitting station detects whether another station has transmitted simultaneously.

### **Basic Operation**

```
Station
|
v
Sense Channel
|
```



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### Important Characteristics

- A transition occurs in the middle of every bit.
  - Provides self-synchronization.
  - Does not contain a DC component.
  - Requires greater bandwidth compared with some other encoding schemes.
- 

## Q8. Explain the working of ARP and RARP. Discuss the role of the MAC Layer in Data Communication.

### 1. Introduction

Communication in a LAN requires both:

- **Logical addressing** — IP address
- **Physical addressing** — MAC address

A device uses an IP address to identify a network-level destination and a MAC address to deliver the frame over the local network.

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## 2. ARP — Address Resolution Protocol

**ARP** resolves an **IPv4 address** to a corresponding **MAC address**.

### Need for ARP

Suppose a host knows the destination IP address:

Destination IP: 192.168.1.20

To construct an Ethernet frame, it also requires the destination MAC address.

ARP performs this mapping.

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### ARP Working

#### Step 1: ARP Cache Lookup

The sender checks its ARP cache.

```
Does IP → MAC mapping exist?  
  |  
Yes / \ No  
  /  \  
Use Entry  Send ARP Request
```

#### Step 2: ARP Request

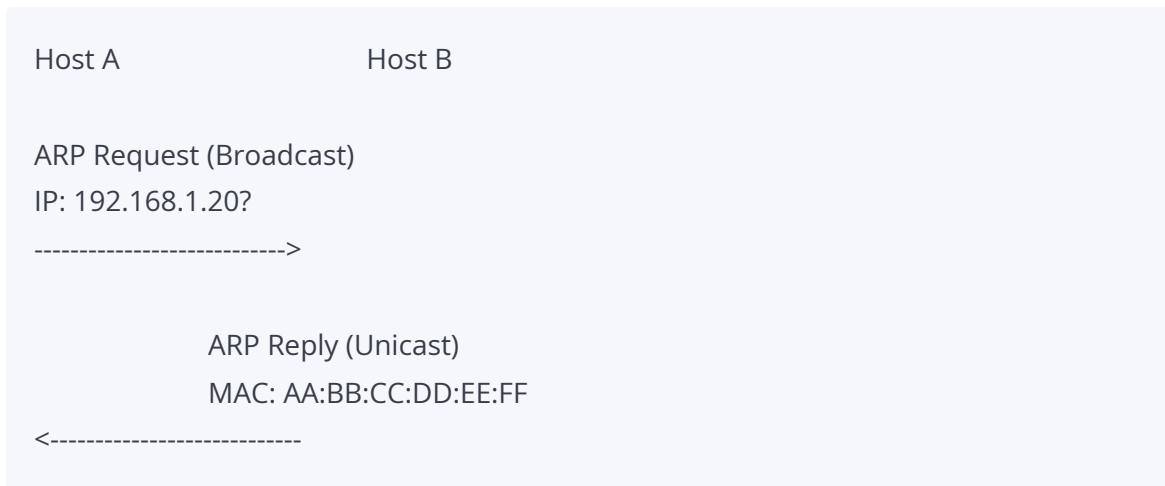
The sender broadcasts an ARP request:

"Who has IP address 192.168.1.20?"

The request is delivered to all hosts in the local broadcast domain.

#### Step 3: ARP Reply

The host with the matching IP address sends a reply containing its MAC address.



#### Step 4: Cache Update

The sender stores the IP-to-MAC mapping in its ARP cache and uses the MAC address to construct the Ethernet frame.

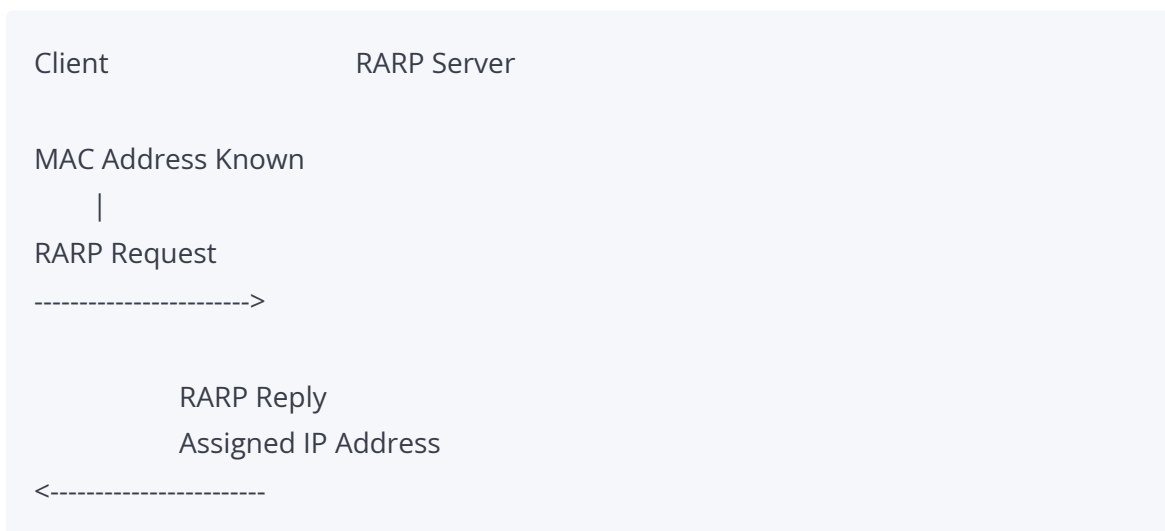
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### 3. RARP — Reverse Address Resolution Protocol

**RARP** performs the reverse mapping:

MAC Address → IP Address

RARP was historically used by devices that knew their hardware address but needed to obtain their IP address from a RARP server.



#### Limitation of RARP

RARP has been largely replaced by protocols such as **BOOTP** and **DHCP**, which provide additional network configuration information.

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## 5. Role of the MAC Layer

The **Media Access Control (MAC) sublayer** is the lower sublayer of the Data Link Layer. It controls how network devices access a shared communication medium.

### Major Functions

#### 1. Physical Addressing

The MAC layer uses a **48-bit MAC address** to identify network interfaces on a local network.

Example:

```
00:1A:2B:3C:4D:5E
```

#### 2. Frame Encapsulation

The MAC layer encapsulates network-layer packets into frames by adding:

- Destination MAC address
- Source MAC address
- Control information
- Frame Check Sequence

#### 3. Medium Access Control

It determines when a device may access the transmission medium.

Examples:

- CSMA/CD in classical Ethernet
- CSMA/CA in wireless LANs

#### 4. Error Detection

**The MAC frame includes an FCS, typically generated using CRC, to detect frame corruption.**

## Q9. Explain Distance Vector Routing in detail.

### 1. Introduction

**Distance Vector Routing (DVR)** is a dynamic routing algorithm in which each router determines the best route to every destination based on:

1. **Distance (metric)** — Cost of reaching the destination.
2. **Vector (next hop)** — Direction or neighboring router through which the destination can be reached.

Each router maintains a **routing table** containing its current best-known distance to every destination and periodically exchanges this information with its immediate neighbors.

Distance Vector Routing is based on the **Bellman-Ford algorithm**.

---

### 2. Routing Table

A typical distance vector routing table contains:

| Destination | Distance/Metric | Next Hop |
|-------------|-----------------|----------|
| Network A   | 0               | —        |
| Network B   | 2               | R2       |
| Network C   | 5               | R3       |

Where:

- **Destination** identifies the target network.
- **Metric** represents the cost of the route.
- **Next Hop** identifies the neighboring router used to forward packets.

The metric may be based on:

- Hop count
  - Delay
  - Bandwidth
  - Administrative cost
-

### 3. Basic Working Principle

Each router initially knows:

- The cost of directly connected networks.
- The cost of neighboring routers.

The cost to unknown destinations is considered:

$$\infty$$

Periodically, each router sends its distance vector to its directly connected neighbors.

When a router receives an update, it calculates whether a better route exists.

The Bellman-Ford update equation is:

$$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$$

Where:

- $(D_x(y))$  = distance from router (x) to destination (y)
- $(c(x,v))$  = cost from router (x) to neighbor (v)
- $(D_v(y))$  = distance from neighbor (v) to destination (y)

The router selects the neighbor that provides the **minimum total path cost**.

---

### 5. Distance Vector Update Procedure

The operation of a router can be summarized as follows:

Receive Distance Vector from Neighbor

|  
v

For each Destination

|  
v

Calculate:

Cost to Neighbor + Neighbor's Distance

|  
v

Is New Cost Smaller?



|                                      |                          |
|--------------------------------------|--------------------------|
| /                                    | \                        |
| Yes                                  | No                       |
|                                      |                          |
| v                                    | v                        |
| Update Routing<br>Table and Next Hop | Retain Existing<br>Route |

This process continues until all routers reach a stable state called **convergence**.

---

## 6. Advantages of Distance Vector Routing

### 1. Simple Implementation

The algorithm is comparatively simple and easy to implement.

### 2. Low Computational Complexity

Each router exchanges information primarily with its immediate neighbors.

### 3. Distributed Operation

No central routing controller is required.

### 4. Dynamic Routing

Routing tables can adapt to topology and link-cost changes.

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## 7. Limitations

### 1. Slow Convergence

The network may take time to converge after a topology change.

### 2. Routing Loops

Incorrect routing information can circulate between routers, causing packets to loop.

### 3. Count-to-Infinity Problem

When a route fails, routers may repeatedly advertise increasing metrics to one another.

Example:

```
A believes route is through B
B believes route is through A

A  Metric 2
B  Metric 3
A  Metric 4
B  Metric 5
...
```

The metric continues increasing until the predefined infinity value is reached.

---

## 8. Solutions to Routing Problems

### Split Horizon

A router does not advertise a route back through the same interface from which that route was learned.

### Route Poisoning

A failed route is advertised with an infinite metric to indicate that the route is unreachable.

### Hold-Down Timer

The router temporarily prevents potentially incorrect updates from replacing a recently failed route.

### Triggered Updates

A routing update is sent immediately after a significant routing change instead of waiting for the regular periodic update.

---

## 9. Distance Vector Routing Protocol: RIP

**Routing Information Protocol (RIP)** is a well-known Distance Vector Routing Protocol.

### Important Characteristics

- Uses **hop count** as its routing metric.
- The maximum valid hop count is **15**.
- A metric of **16** represents infinity or an unreachable destination.
- Routers periodically exchange routing information with their neighbors.

Because of its hop-count limitation, RIP is more suitable for relatively small networks.

---

## Distance Vector Routing vs Link State Routing

| Parameter             | Distance Vector               | Link State                 |
|-----------------------|-------------------------------|----------------------------|
| Information exchanged | Routing table/distance vector | Link-state information     |
| Algorithm             | Bellman-Ford                  | Dijkstra/SPF               |
| Network knowledge     | Neighbor-based                | Complete topology database |
| Convergence           | Generally slower              | Generally faster           |
| Resource requirement  | Lower                         | Higher                     |
| Complexity            | Lower                         | Higher                     |
| Example               | RIP                           | OSPF                       |

---